

Dose escalation and expansion of nivolumab plus relatlimab in solid tumors in RELATIVITY-020 Parts A-C

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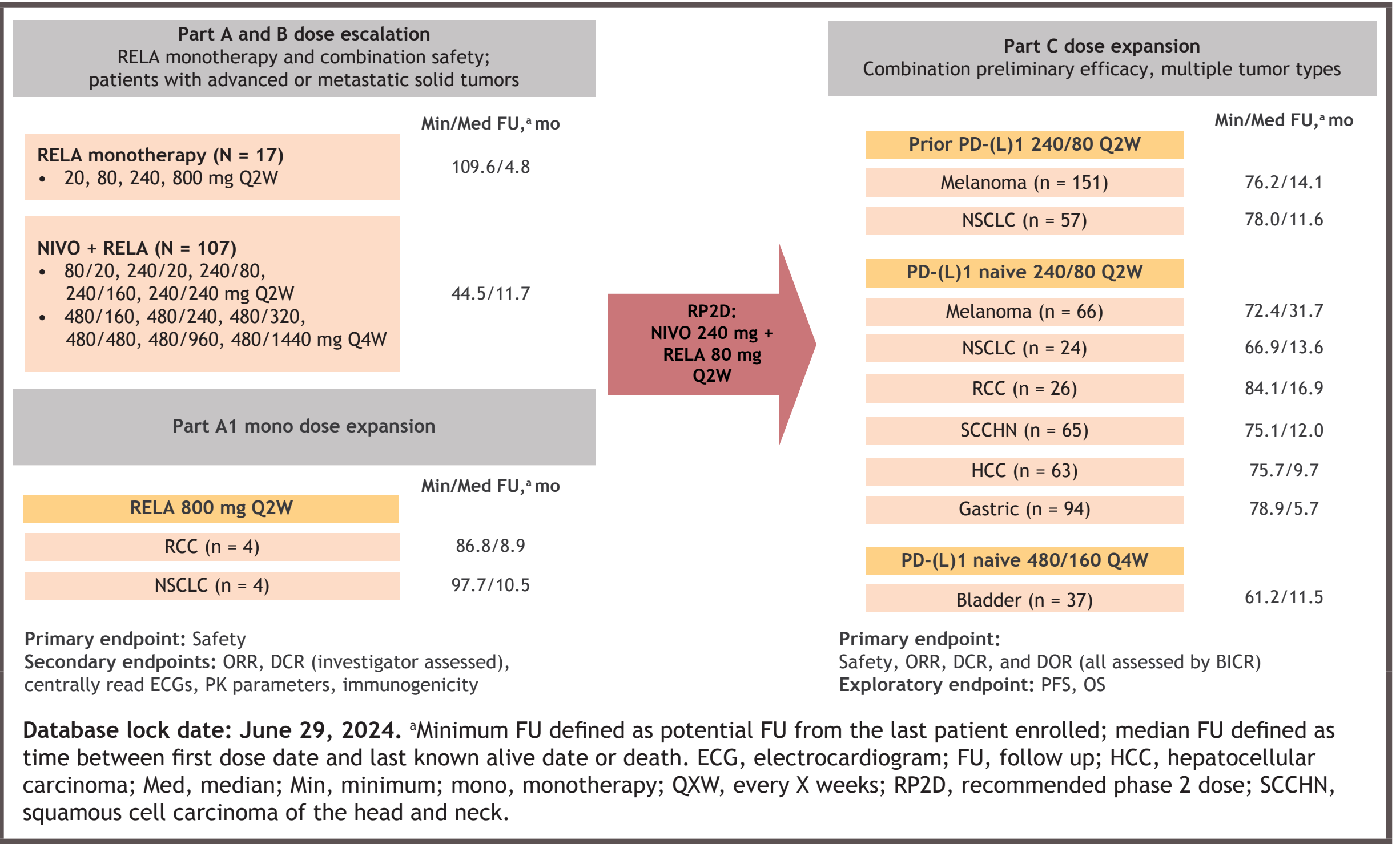
Introduction

- On the basis of phase 2/3 **RELATIVITY-047** results, nivolumab + relatlimab (NIVO + RELA) is the only anti-programmed death-1 (PD-1) + anti-lymphocyte-activation gene 3 (LAG-3) combination approved for the treatment of patients with advanced melanoma^{1,3}
 - NIVO + RELA demonstrated a significant progression-free survival (PFS) benefit per blinded central independent review (BICR) in comparison with NIVO alone at the primary analysis (median follow-up, 13.2 months): hazard ratio, 0.75 (95% CI, 0.62-0.92; P = 0.006)¹
 - Long-term follow up at a median of 34.9 months confirmed a benefit in overall survival (OS)³
- The **RELATIVITY-020** trial was a 5-part, phase 1/2a dose-escalation and cohort-expansion open-label trial designed to assess the safety, tolerability, and effectiveness of an anti-LAG-3 with and without an anti-PD-1 in the treatment of advanced solid tumors (NCT01968109)
- Here we report safety, efficacy, and pharmacokinetic (PK) results from Parts A-C of the study

Methods

- RELATIVITY-020** was designed with 5 parts; Parts A-C are presented here (**Figure 1**)
 - Part A: dose-escalation analysis in patients with immuno-oncology (IO) treatment-naïve solid tumors treated with RELA monotherapy at 20-800 mg
 - Part A1: expansion studies in patients with IO-refractory non-small cell lung cancer (NSCLC) and renal cell carcinoma (RCC)
 - Part B: dose-escalation across multiple solid tumor types with NIVO + RELA combination from 80/20 mg to 240/240 mg Q2W or 480/160 mg to 480/1440 mg Q4W
 - Part C: expansion in solid tumors at NIVO + RELA 240/80 mg Q2W (with bladder at 480/160 mg Q4W)
 - Parts D and E assessed safety and tolerability of NIVO + RELA in patients with advanced melanoma who were refractory to anti-PD-1/anti-programmed death ligand 1 (PD-L1) therapy (Part D)⁴ and/or in patients with advanced melanoma who were treatment-naïve for anti-PD-1 treatment (Part E)⁵
- Primary endpoint was safety (rate of adverse events [AEs], serious AEs [SAEs], AEs leading to treatment discontinuation, deaths, laboratory abnormalities) in Parts A, B, and C
 - Dose-limiting toxicities were used to determine maximum tolerated dose (MTD) in Parts A and B, and no MTD was observed
- Primary endpoints included BICR-assessed overall response rate (ORR), disease control rate (DCR), and duration of response (DOR) in Part C
- Secondary endpoints included PK, immunogenicity, and investigator-assessed ORR and DCR

Figure 1. **RELATIVITY-020** study design (Parts A-C)



Results

- Baseline demographics and disease characteristics were similar across Parts A-C (**Table 1**)
 - Most patients were from North America in Parts A and B and from Europe in Part C
 - Normal LDH levels were observed in 44.0% of patients from Part A, 60.7% from Part B, and 22.7% from Part C
 - LAG-3 expression was comparable across cohorts
- In Part C, baseline characteristics were similar among patients with melanoma and those with NSCLC, regardless of prior exposure to IO therapy (**Table 2**)
 - Prior radiotherapy was observed in 41.7% of patients with IO-naïve NSCLC, 66.7% of patients with NSCLC that progressed on IO, 18.2% of patients with IO-naïve melanoma, and 33.1% of patients with melanoma that progressed on IO
- At the time of the database lock of June 29, 2024, of the 715 patients across Parts A-C, there were 114 patients continuing in the study and 563 total deaths (**Table 3**)
- Across Parts A-C, NIVO + RELA dose escalation demonstrated an acceptable safety profile, with no MTD identified
 - Grade 3/4 TRAEs occurred in approximately 16% of patients across Parts A-C; any-grade TRAEs led to discontinuation of treatment in approximately 10%-14% of patients overall (4%-11% for grade 3/4) (**Table 4**)
 - Rates of grade 3/4 TRAEs were higher in IO-naïve patients than in those previously treated with IO (NSCLC, 16.7% vs 10.5%; melanoma, 24.2% vs 11.9%)
 - Most common any-grade immune-mediated adverse events (IMAEs) in Part C were diarrhea/colitis, rash, hypothyroidism/thyroiditis, and hyperthyroidism (**Table 5**)

Table 1. Baseline demographics and disease characteristics

	Part A/A1 RELA mono dose escalation and expansion (N = 25)	Part B NIVO + RELA dose escalation (N = 107)	Part C pooled Q2W* (N = 546)	Part C Q4W* (N = 37)
Median age, years (range)	59.0 (34-78)	59.0 (23-74)	61.0 (18-88)	68.0 (54-79)
Male, n (%)	16 (64.0)	60 (56.1)	362 (66.3)	26 (70.3)
Europe/North America, n (%)	4 (16.0)/21 (84.0)	24 (22.4)/83 (77.6)	480 (87.9)/66 (12.1)	35 (94.6)/2 (5.4)
Presence of liver metastases, n (%)	12 (48.0)	37 (34.6)	190 (34.8)	11 (29.7)
ECOG PS, ^a n (%)				
0	13 (52.0)	61 (57.0)	313 (57.3)	22 (59.5)
1/2	12 (48.0)/0	45 (42.1)/1 (0.9)	220 (40.3)/1 (0.2)	14 (37.8)/1 (2.7)
Brain metastases, ^a n (%)				
Yes	0	3 (2.8)	16 (2.9)	0
No/unknown	1 (4.0)/24 (96.0)	19 (17.8)/85 (79.4)	188 (34.4)/342 (62.6)	0/37 (100)
LDH, ^a n (%)				
Normal	11 (44.0)	65 (60.7)	124 (22.7)	NA
Normal to < 2x ULN ≥ 2x ULN	5 (20.0)/1 (4.0)	29 (27.1)/11 (10.3)	53 (9.7)/34 (6.2)	NA
Unknown/not applicable	0/8 (32.0)	2 (1.9)/0	6 (1.1)/0	NA
LAG-3 status 1% cutoff, n (%)				
Positive	9 (36.0)	30 (28.0)	226 (41.4)	9 (24.3)
Negative/unevaluable	13 (52.0)/3 (12.0)	59 (55.1)/18 (16.8)	241 (44.1)/79 (14.5)	16 (43.2)/12 (32.4)

*Part C pooled Q2W includes tumor cohorts in which 240 mg NIVO + 80 mg RELA Q2W dose and schedule were studied. This includes IO-naïve cohorts in patients with melanoma, NSCLC, RCC, SCCHN, HCC, and gastric cancer, and cohorts of patients with melanoma and NSCLC who progressed on IO. *Part C Q4W includes cohort of patients with bladder cancer who were previously untreated with IO in which 480 mg NIVO + 160 mg RELA Q4W dose and schedule were studied. *ECOG PS was not reported for 11 patients with RCC and 1 patient with gastric cancer. For patients with RCC, ECOG-PS was not required; Karnofsky Performance Score was 100 for 10 patients (88.5%), 90 for 11 patients (42.3%), and 80 for 3 patients (19.2%). *Only available for melanoma patients in Part C. ECOG PS, Eastern Cooperative Oncology Group performance status; LDH, lactate dehydrogenase; NA, not applicable; ULN, upper limit of normal.

Table 2. Baseline demographics and disease characteristics of patients with IO-naïve and IO-progressed NSCLC and melanoma (Part C)

	NSCLC IO naïve (n = 24)	NSCLC IO progressed (n = 57)	Melanoma IO naïve (n = 66)	Melanoma IO progressed (n = 151)
Median age, years (range)	63.0 (42-88)	63.0 (40-78)	64.5 (24-87)	60.0 (25-81)
Male, n (%)	16 (66.7)	37 (64.9)	36 (54.5)	92 (60.9)
Race, ^a n (%)				
White	23 (95.8)	55 (96.5)	66 (100)	147 (97.4)
Black/Asian	0	0	0	1 (0.7)/1 (0.7)
ECOG PS, n (%)				
0/1 ^a	11 (45.8)/13 (54.2)	31 (54.4)/26 (45.6)	55 (83.3)/11 (16.7)	104 (68.9)/47 (31.1)
Stage III/Stage IV, n (%)	0/19 (79.2)	0/46 (80.7)	5 (7.6)/61 (92.4)	7 (4.6)/144 (95.4)
Presence of liver metastases, n (%)	2 (8.3)	9 (15.8)	19 (28.8)	53 (35.1)
Prior surgery, n (%)	9 (37.5)	27 (47.4)	63 (95.5)	140 (92.7)
Prior radiotherapy, n (%)	10 (41.7)	38 (66.7)	12 (18.2)	50 (33.1)
Prior systemic therapy, n (%)	10 (41.7)	57 (100)	10 (15.2)	151 (100)
Immunotherapy, n (%)	1 (4.2)	57 (100)	9 (13.6)	151 (100)
Chemotherapy, n (%)	10 (41.7)	57 (100)	2 (3.0)	39 (25.8)
No. of prior systemic regimens, n (%)				
1	7 (29.2)	1 (1.8)	9 (13.6)	45 (29.8)
2	2 (8.3)	25 (43.9)	1 (1.5)	42 (27.8)
≥ 3-5	1 (4.2)	31 (54.4)	0	64 (42.4)
Median (range)	1.0 (1-4)	3.0 (1-5)	1.0 (1-2)	2.0 (1-9)

^aOther (n [23]): NSCLC IO naïve, 1 (4.2); NSCLC IO progressed, 2 (3.5); melanoma IO naïve, 0; melanoma IO progressed, 2 (1.3).

Table 3. Patient disposition

	Part A/A1 RELA mono dose escalation and expansion (N = 25)	Part B pooled NIVO + RELA dose escalation (N = 107)	Part C pooled Q2W* (N = 546)	Part C Q4W* (N = 37)
Continuing the treatment period, n (%)	0	2 (1.9)	1 (0.2)	0
Not continuing the treatment period, n (%)				
Disease progression	25 (100)	105 (98.1)	545 (99.8)	37 (100)
Study drug toxicity	20 (80.0)	79 (73.8)	416 (76.2)	25 (67.6)
AE unrelated to study treatment	3 (12.0)	12 (11.2)	48 (8.8)	4 (10.8)
Patient request/withdraw consent	0	5 (4.7)	20 (3.7)	3 (8.1)
Maximum clinical benefit	1 (4.0)/0	3 (2.8)/2 (1.9)	14 (2.6)/3 (0.5)	0/1 (2.7)
No longer met study criteria	0	0	9 (1.6)	0
Completed treatment per protocol	0	4 (0.7)	0	0
Other	1 (4.0)	2 (1.9)	29 (5.3)	1 (2.7)
Continuing in the study, n (%)	2 (8.0)	19 (17.8)	88 (16.1)	5 (13.5)
Not continuing in the study, n (%)				
Patient withdrew consent	23 (92.0)	88 (82.2)	458 (83.9)	32 (86.5)
Lost to follow-up	1 (4.0)	5 (4.7)	12 (2.2)	1 (2.7)
Death	0	0	5 (0.9)	0
Follow-up no longer required per protocol	17 (68.0)	33 (30.8)	218 (39.9)	13 (35.1)
Other/not reported	0/5 (20.0)	2 (1.9)	11 (2.0)	0
Deaths				
Disease	23 (92.0)	84 (78.5)	427 (78.2)	29 (78.4)
Study drug toxicity	22 (88.0)	75 (70.1)	388 (71.1)	29 (78.4)
Unknown/other	0	1 (0.9) ^a	3 (0.5) ^a	0
	1 (4.0)/0	6 (5.6)/2 (1.9) ^a	18 (3.3)/18 (3.3) ^a	0/0

*Part C pooled Q2W includes tumor cohorts in which 240 mg NIVO + 80 mg RELA Q2W dose and schedule were studied. *Part C Q4W includes cohort of patients with bladder cancer who were previously untreated with IO in which 480 mg NIVO + 160 mg RELA Q4W dose and schedule were studied. *The cause of death per investigator was myocarditis. *The causes of death per investigator were increased dyspnea, immune-mediated pneumonitis, and pulmonary fibrosis. *The verbatim terms reported for the "other" reasons for death were respiratory failure (RELA + NIVO 240/80 mg Q4W) and COVID-19 infection (RELA + NIVO 1440/480 mg Q4W). *The verbatim terms reported for the "other" reasons for death were acute heart attack, CMV reactivation with pneumonia complicated by hemophagocytic lymphohistiocytosis, sepsis, pneumonia (n = 3), respiratory failure, septic shock, disease progression, aspiration, sepsis with percutaneous endoscopic gastrostomy infection differential diagnosis pneumonia, respiratory distress syndrome following bacteremic infectious pneumonitis, Parkinson's/PSP, acute hypoxic hyperbaric respiratory failure caused by aspiration secondary to dysphagia, multiple organ failure, cardiac arrest, assisted suicide leading to death, and hemorrhagic cerebellar hemisphere sinister. CMV, cytomegalovirus; PSP, progressive supranuclear palsy.

Table 4. Safety summary

	Part A/A1 RELA mono dose escalation and expansion (N = 25)	Part B pooled NIVO + RELA dose escalation (N = 107)	Part C pooled Q2W* (N = 546)	Part C Q4W* (N = 37)
Any grade				
Grade 3/4				
AEs, n (%)	25 (100)	5 (20.0)	106 (99.1)	44 (41.1)
TRAEs, n (%)	15 (60.0)	4 (16.0)	77 (72.0)	17 (15.9)
SAEs, n (%)	18 (72.0)	4 (16.0)	72 (67.3)	36 (33.6)
Treatment-related SAEs, n (%)	2 (8.0)	1 (4.0)	16 (15.0)	11 (10.3)
AEs leading to discontinuation, n (%)	6 (24.0)	1 (4.0)	21 (19.6)	13 (12.1)
TRAEs leading to discontinuation, n (%)	3 (12.0)	1 (4.0)	13 (12.1)	9 (8.4)

*Part C pooled Q2W includes tumor cohorts in which 240 mg NIVO + 80 mg RELA Q2W dose and schedule were studied. *Part C Q4W includes cohort of patients with bladder cancer who were previously untreated with IO in which 480 mg NIVO + 160 mg RELA Q4W dose and schedule were studied.

Table 5. Any-grade IMAEs in patients from Part C

	Part C pooled Q2W* (N = 546)	Part C Q4W* (N = 37)
Any grade		
IMAEs treated with IMM, n (%)		
Pneumonitis	19 (3.5)	1 (2.7)
Diarrhea/colitis	30 (5.5)	3 (8.1)
Hepatitis	22 (4.0)	1 (2.7)
Nephritis/renal dysfunction	6 (1.1)	1 (2.7)
Rash	47 (8.6)	4 (10.8)
Hypersensitivity/infusion reactions	1 (0.2)	1 (2.7)
Endocrine IMAEs treated with or without IMM, n (%)		
Adrenal insufficiency	9 (1.6)	0
Hypothyroidism/thyroiditis	49 (9.0)	3 (8.1)
Hypothyroidism	45 (8.2)	3 (8.1)
Thyroiditis	6 (1.1)	0
Diabetes mellitus	4 (0.7)	0
Hyperthyroidism	26 (4.8)	2 (5.4)
Hypophysitis	11 (2.0)	1 (2.7)

*Part C pooled Q2W includes tumor cohorts in which 240 mg NIVO + 80 mg RELA Q2W dose and schedule were studied. *Part C Q4W includes cohort of patients with bladder cancer who were previously untreated with IO in which 480 mg NIVO + 160 mg RELA Q4W dose and schedule were studied. IMM, immune-modulating medication.

- In Part C of the study, efficacy was noted across several tumor types tested, including in patients with melanoma and NSCLC who had progressed on prior IO therapy (**Table 6**)
- As was expected, for both patients with melanoma and those with NSCLC, improvements in PFS and OS were greater in IO-naïve patients than in those previously treated with IO (**Figure 2**)
 - However, differences could likely be reflected by baseline prognostic differences rather than by treatment effect
- LAG-3 expression enriched for efficacy in patients with melanoma and NSCLC, especially in IO-naïve patients (**Figure 2**)

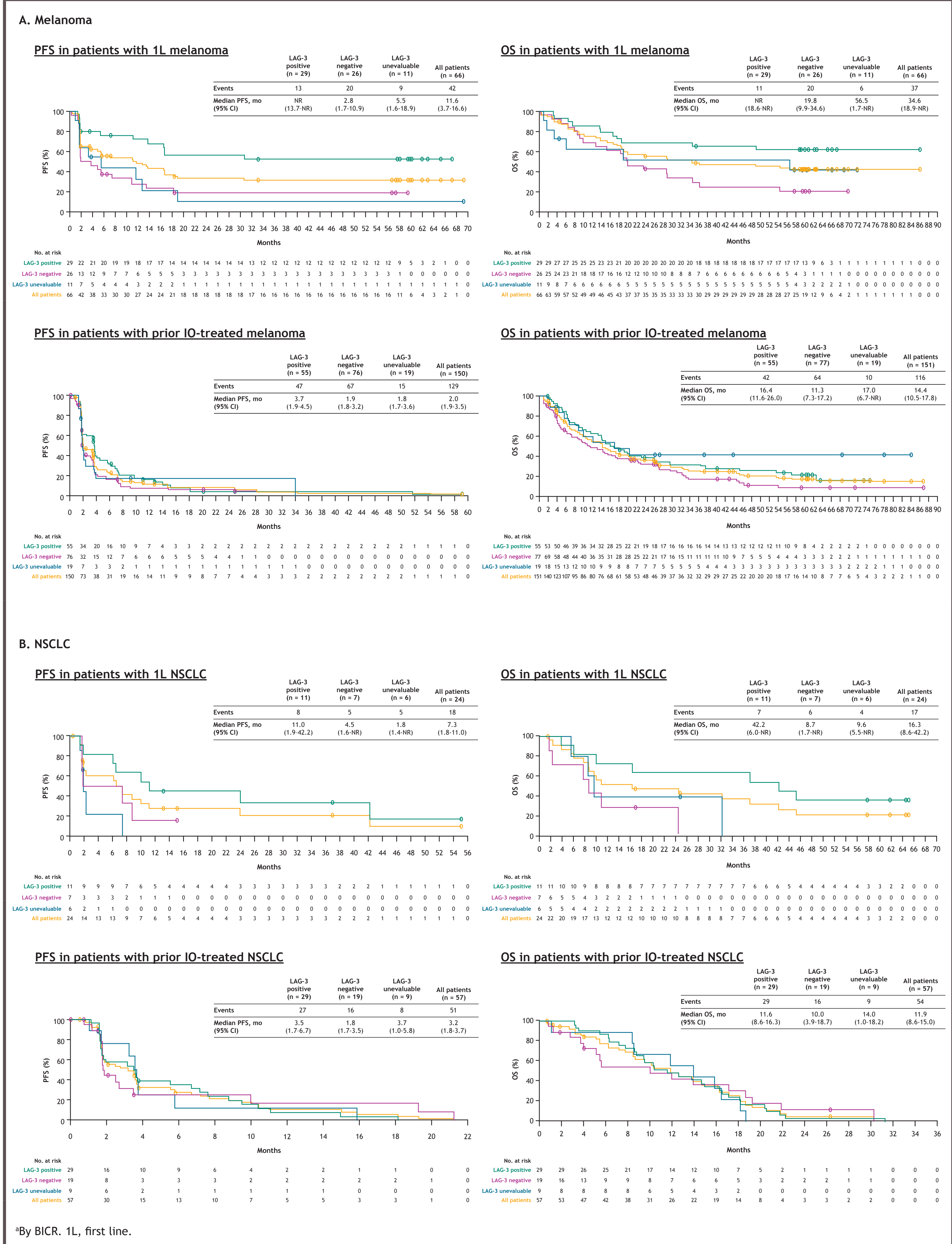
Table 6. Part C efficacy summary

	IO naïve						IO progressed		
	Melanoma (n = 66)	NSCLC (n = 24)	RCC (n = 26)	SCCHN (n = 65)	HCC (n = 63)	Gastric (n = 93)	Bladder (n = 36)	Melanoma (n = 150)	NSCLC (n = 57)
ORR per BICR, % (95% CI)	47.0 (34.6-59.7)	29.2 (12.6-51.1)	15.4 (4.4-34.9)	15.4 (7.6-26.5)	15.9 (7.9-27.3)	8.6 (3.8-16.2)	30.6 (16.3-48.1)	10.7 (6.2-16.7)	5.3 (1.1-14.6)
Median DOR per BICR, months (95% CI)	NR (11.1-NR)	40.4 (6.4-NR)	NR (NR)	27.9 (3.7-NR)	54.6 (13-NR)	35.1 (11.2-NR)	18.7 (6.9-NR)	21.1 (4.0-46.1)	8.6 (5.6-NR)
12-week DCR per BICR, % (95% CI)	59.1 (46.3-71.0)	50.0 (29.1-70.9)	46.2 (26.6-66.6)	49.2 (36.6-61.9)	50.8 (37.9-63.6)	25.8 (17.3-35.9)	44.4 (27.9-61.9)	38.0 (20.4-46.3)	40.4 (27.6-54.2)
Median PFS, months (95% CI)	11.6 (3.7-16.6)	7.3 (1.8-11.0)	3.4 (1.6-5.6)	3.6 (1.9-4.1)	3.5 (1.9-3.8)	1.8 (1.7-1.9)	3.1 (1.8-8.9)	2.0 (1.9-3.5)	3.2 (1.8-3.7)
12-month PFS rate, % (95% CI)	48 (35-60)	28 (11-47)	NA	17 (8-27)	24 (14-36)	9 (4-16)	25 (12-41)	13 (8-20)	11 (4-21)
Median OS, ^a months (95% CI)	34.6 (18.9-NR)	16.3 (8.6-42.2)	17.2 (9.0-43.8)	12.0 (7.2-17.5)	10.3 (8.3-15.3)	6.1 (4.9-9.0)	12.1 (5.8-26.0)	14.4 (10.5-17.8)	11.9 (8.6-15.0)
12-month OS rate, ^a % (95% CI)	76 (63-84)	52 (31-70)	61 (39-77)	49 (36-60)	46 (33-58)	29 (20-39)	53 (35-67)	55 (47-63)	47 (34-60)

^aFor OS, N = 151 for patients with melanoma that progressed on IO (includes all treated patients). NR, not reached.

- PK and immunogenicity testing for NIVO + RELA showed the following:
 - RELA exposure increased proportionally to dose when given as monotherapy or in combination with NIVO across the studied dose ranges
 - Steady state concentration of RELA was reached by 16 weeks, and systemic accumulation ratio was approximately 2- to 3-fold following NIVO + RELA 240/80 mg Q2W
 - RELA exposures following administration of RELA 80 mg Q2W and NIVO + RELA 240/80 mg Q2W were comparable, indicating no PK interaction between RELA and NIVO
 - There were no incidences of RELA antidrug antibodies (ADA) in patients treated with RELA monotherapy in Part A and no apparent increases in RELA or NIVO ADA with increasing doses of RELA in combination with NIVO in Part B
 - The incidence rates of RELA and NIVO ADA and neutralizing antibody (NAB) were similar between melanoma and nonmelanoma patients treated with NIVO + RELA single-agent vs 240/80 mg Q2W in Part C, and there were no apparent trends showing an effect of ADA or NAB on safety or efficacy
 - The average exposure (C_{avg}) is similar between 240/80 mg Q2W and 480/160 mg Q4W⁶

Figure 2. PFS^a and OS in patients with melanoma and NSCLC



Summary

- In the **RELATIVITY-020** study, NIVO + RELA had an acceptable safety profile at all dose levels, consistent with the profiles of the individual components
 - No MTD was identified up to the NIVO + RELA 480/1440 mg Q4W dosing regimen in Part B
- NIVO + RELA demonstrated preliminary efficacy across several different tumor types
 - Clinical activity was observed in both patients with IO-naïve and IO-refractory disease
- PK data reaffirmed the combination of NIVO + RELA 480/160 mg to be most efficacious in melanoma
- These results support continued investigation of NIVO + RELA in the treatment of patients with solid tumors

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